# Which retrieval settings find the true value?

Truth coverage and graded relevance of the documents medsim retrieves (Europe PMC + LitSense), on the 141 set A questions of the retrieval benchmark · judge anthropic/claude-haiku-5.5 · 37 scored configurations and controls

> **Result.** The best configuration is **llm_select20+epmc25+q_synonym** (an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion): strict truth coverage 70.2% against the baseline's 60.3% (+9.9 (+3.5 to +17.0) points, paired over the same questions), nDCG@8 0.750 against 0.691, and precision@8 49.6% against 41.3%. On fresh queries: query set fresh1: strict +7.1 points, nDCG +0.054; query set fresh2: strict +12.1 points, nDCG +0.058.

## What was measured

- **Strict truth coverage** (decides): a document gives a value or range for this variable in patients with this or a closely related condition, and the hidden true value lies inside it or within about 15%. **Category coverage** (shown): a document's value, range, or level, a reference range for comparable patients included, puts the true value in the same low / normal / high category.
- **Graded relevance**: each document 0 (not about the variable or the condition), 1 (about them, no usable value), or 2 (a value for the variable in comparable patients). **nDCG@8** is against a perfect list of eight grade-2 documents, so it rewards better documents and a better order (it breaks ties); **precision@8** is the share of grade-2 documents.
- One judge call per question grades all eight documents and gives both coverage verdicts. Every positive verdict needs a verbatim quote that code finds in the named document; an unverified grade 2 counts as 1, and unverified coverage as not covered.
- Each configuration retrieves for the same **frozen search query** per question (medsim's Stages A and B once, with the stored runs' model deepseek/deepseek-v4-flash-0731), and no answer is written. Differences between configurations therefore come from the retrieval settings. Eight documents per question throughout. Search results are cached, so a configuration that issues the same requests gets the same results.

## Can the metrics be trusted?

| Check | Strict coverage | Category | nDCG@8 | Reading |
|---|---|---|---|---|
| Shuffled control: each question judged on another question's documents | 0.0% | 0.7% | 0.034 | Near zero: the judge reads the documents rather than answering from the question. |
| The baseline's documents judged twice | 60.3% / 61.7% | 83.7% / 85.8% | 0.691 / 0.680 | The judge's own noise: 1.4 points of strict coverage. |
| Stored benchmark runs 2–6 of the baseline method (fresh queries, full pipeline) | 61.7% ± 3.4 | – | 0.675 ± 0.019 | Run-to-run spread (mean ± SD) when the query is written anew; the frozen-query baseline (60.3%) lies inside it. |
| Stored run 2 of the original retrieval (`current`) | 40.4% | 70.2% | 0.593 | About 20 points below changes 1–4: the metrics separate a known difference. |

## Single changes from the baseline

The baseline is medsim's default retrieval (changes 1–4: value-first ranking in one list, 50 Europe PMC candidates, animal and age filter, article-body search with excerpts, value-based broadening). Each row changes one setting. Differences are paired over the same 141 questions, with 95% bootstrap intervals; green intervals exclude zero.

| Configuration | Strict coverage | Δ strict vs baseline (95% CI) | Category coverage | nDCG@8 | Δ nDCG vs baseline | Precision@8 | n |
|---|---|---|---|---|---|---|---|
| **baseline**<br>baseline: changes 1-4 | 60.3% | | 83.7% | 0.691 | | 41.3% | 141 |
| **llm_select40**<br>an LLM picks the 8 documents from the best 40 | 68.8% | +8.5 (+2.1 to +15.6) | 89.4% | 0.746 | +0.055 (+0.030 to +0.078) | 50.8% | 141 |
| **llm_select20**<br>an LLM picks the 8 documents from the best 20 | 67.4% | +7.1 (+0.7 to +14.2) | 89.4% | 0.749 | +0.058 (+0.041 to +0.076) | 49.6% | 141 |
| **epmc25**<br>25 Europe PMC candidates (baseline 50) | 66.0% | +5.7 (-1.4 to +12.8) | 85.1% | 0.690 | -0.001 (-0.020 to +0.017) | 42.6% | 141 |
| **q_synonym**<br>Europe PMC MeSH synonym expansion | 63.8% | +3.5 (-1.4 to +8.5) | 84.4% | 0.690 | -0.001 (-0.015 to +0.013) | 42.3% | 141 |
| **epmc_only**<br>Europe PMC only | 63.1% | +2.8 (-2.8 to +8.5) | 85.8% | 0.665 | -0.026 (-0.048 to -0.005) | 40.5% | 141 |
| **ls_sentences**<br>LitSense sentences instead of passages | 63.1% | +2.8 (-4.3 to +9.9) | 86.5% | 0.659 | -0.032 (-0.053 to -0.013) | 39.3% | 141 |
| **q_natural**<br>LitSense natural-language queries | 62.4% | +2.1 (-4.3 to +8.5) | 83.7% | 0.684 | -0.007 (-0.029 to +0.014) | 41.3% | 141 |
| **round_robin**<br>round-robin merge across sources instead of one list | 62.4% | +2.1 (-4.3 to +8.5) | 83.7% | 0.667 | -0.024 (-0.044 to -0.006) | 39.6% | 141 |
| **ls100**<br>100 LitSense results | 61.7% | +1.4 (-5.0 to +7.8) | 86.5% | 0.681 | -0.010 (-0.028 to +0.007) | 40.9% | 141 |
| **epmc100**<br>100 Europe PMC candidates | 61.7% | +1.4 (-5.0 to +7.8) | 85.8% | 0.679 | -0.012 (-0.028 to +0.004) | 40.5% | 141 |
| **fulltext20**<br>20 open-access full texts fetched for excerpts (baseline 10) | 61.0% | +0.7 (-5.7 to +7.1) | 87.9% | 0.700 | +0.009 (-0.008 to +0.026) | 44.7% | 141 |
| **ls60**<br>60 LitSense results (baseline 30) | 61.0% | +0.7 (-5.7 to +7.1) | 86.5% | 0.681 | -0.010 (-0.025 to +0.005) | 41.2% | 141 |
| **ls_no_rerank**<br>no LitSense-side rerank | 61.0% | +0.7 (-5.7 to +7.1) | 87.2% | 0.677 | -0.014 (-0.033 to +0.004) | 40.5% | 141 |
| **chars3000**<br>3,000 characters per document (baseline 1,500) | 60.3% | +0.0 (-6.4 to +6.4) | 86.5% | 0.677 | -0.013 (-0.035 to +0.006) | 41.7% | 141 |
| **no_value_first**<br>no value-first ranking | 59.6% | -0.7 (-7.1 to +5.7) | 84.4% | 0.683 | -0.008 (-0.026 to +0.009) | 41.5% | 141 |
| **ladder_v1**<br>the original broadening ladder (v1) | 59.6% | -0.7 (-7.8 to +6.4) | 85.8% | 0.661 | -0.029 (-0.050 to -0.010) | 37.8% | 141 |
| **q_gpt6luna**<br>openai/gpt-6-luna writes the search query | 58.9% | -1.4 (-10.6 to +7.1) | 87.9% | 0.704 | +0.014 (-0.016 to +0.040) | 44.0% | 141 |
| **no_population_filter**<br>no animal and age-group filter | 56.7% | -3.5 (-9.2 to +2.1) | 85.8% | 0.682 | -0.008 (-0.026 to +0.008) | 40.7% | 141 |
| **no_excerpts**<br>no full-text excerpts (abstracts and passages only) | 39.0% | -21.3 (-29.8 to -12.8) | 70.9% | 0.597 | -0.094 (-0.122 to -0.068) | 29.7% | 141 |
| **no_lexical_rerank**<br>no lexical rerank within each source | 34.8% | -25.5 (-34.0 to -17.0) | 66.0% | 0.554 | -0.136 (-0.164 to -0.110) | 20.3% | 141 |
| **litsense_only**<br>LitSense only | 29.8% | -30.5 (-40.4 to -21.3) | 61.7% | 0.576 | -0.115 (-0.144 to -0.087) | 23.8% | 141 |

With 21 changes tested at 95%, about one could look better or worse by chance alone; treat single changes whose interval only just excludes zero with care.

## Combinations

The changes with the largest positive estimates, added to the best single change.

| Configuration | Strict coverage | Δ strict vs baseline (95% CI) | Category coverage | nDCG@8 | Δ nDCG vs baseline | Precision@8 | n |
|---|---|---|---|---|---|---|---|
| **baseline**<br>baseline: changes 1-4 | 60.3% | | 83.7% | 0.691 | | 41.3% | 141 |
| **llm_select20+epmc25+q_synonym**<br>**an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion** | **70.2%** | **+9.9** (+3.5 to +17.0) | **87.9%** | **0.750** | **+0.059** (+0.042 to +0.076) | **49.6%** | **141** |
| **llm_select20+q_synonym**<br>an LLM picks the 8 documents from the best 20; Europe PMC MeSH synonym expansion | 68.8% | +8.5 (+2.8 to +14.9) | 90.8% | 0.743 | +0.052 (+0.035 to +0.070) | 48.8% | 141 |
| **llm_select20+epmc25**<br>an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50) | 68.8% | +8.5 (+1.4 to +14.9) | 87.9% | 0.737 | +0.046 (+0.027 to +0.065) | 48.3% | 141 |

## Confirmation on fresh queries

The baseline and the best configuration again, each on two new sets of queries written by the same query builder: the gain should hold when the queries change. Here each Δ compares with the baseline on the same query set.

| Configuration | Strict coverage | Δ strict vs baseline (95% CI) | Category coverage | nDCG@8 | Δ nDCG vs baseline | Precision@8 | n |
|---|---|---|---|---|---|---|---|
| **baseline@fresh1**<br>baseline: changes 1-4 (queries: fresh1) | 62.4% | | 87.9% | 0.681 | | 40.7% | 141 |
| **llm_select20+epmc25+q_synonym@fresh1**<br>**an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion (queries: fresh1)** | **69.5%** | **+7.1** (-0.7 to +14.9) | **89.4%** | **0.735** | **+0.054** (+0.036 to +0.074) | **47.1%** | **141** |
| **baseline@fresh2**<br>baseline: changes 1-4 (queries: fresh2) | 57.4% | | 85.1% | 0.682 | | 41.7% | 141 |
| **llm_select20+epmc25+q_synonym@fresh2**<br>**an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion (queries: fresh2)** | **69.5%** | **+12.1** (+4.3 to +19.9) | **92.9%** | **0.740** | **+0.058** (+0.039 to +0.078) | **47.5%** | **141** |

## Faithfulness of the answers written on these documents

medsim's answer-writing stage (Stage C) was run on each configuration's documents, with the frozen queries, and every answer with documents was judged for faithfulness with the benchmark's rubric (`bench.answer_metrics`): does at least one document give a value for this variable in a comparable patient that agrees with the answer? Same judge; a question with no answer counts as not supported.

| Configuration | Supported (95% CI) | Δ vs baseline (paired) | Covered and supported | Covered, not supported | Not covered, supported | Neither |
|---|---|---|---|---|---|---|
| **baseline** | 22.0% (15.6 to 29.1) | | 23 | 62 | 8 | 48 |
| **llm_select20+epmc25+q_synonym** | **32.6%** (25.5 to 40.4) | **+10.6** (+2.8 to +19.1) | **35** | **64** | **11** | **31** |

"Covered" is strict truth coverage of the same documents. Covered but not supported means the documents held a value matching the truth but the answer was not backed by them: the answer-writing stage, not retrieval, is where those questions are lost. This judge reads "agrees" strictly (in one example, 9.8 cm in the document against 9.6 cm in the answer was not supported).

## Cost and limits

- Cost of this work from the stored usage records: $9.69 (judge $7.15, LLM selection during retrieval $1.79, query writing $0.32, answer writing for faithfulness $0.43). Europe PMC and LitSense are free.
- One judge, an LLM, without human labels: absolute levels may be off; the comparisons between configurations, made by the same judge on the same questions, are the point.
- 141 questions: differences of a few points are inside the noise.

- Europe PMC answered HTTP 503 during the first retrieval of the two fresh-query baseline runs (129 and 64 of 141 questions lost it). Those questions were retrieved again once it recovered and judged again; the tables use the second attempt. Every other run had no failed source.
- LLM selection adds one call per question to medsim's retrieval (about $0.0016 with deepseek/deepseek-v4-flash-0731, and its latency). It is not medsim's default yet.
- The best configuration was chosen on these questions; the fresh-query runs re-test it, but on the same 141 questions, not on new ones.